A Negative Answer to the Universal-Function Version of Erdős's Third Question in Problem 514

April 28, 2026

Abstract

Let f be a transcendental entire function and let

$$M_f(r) = \max_{|z|=r} |f(z)|.$$

Erdős asked, among other things, whether there exists a path to infinity along which $|f(z)|$ tends to infinity faster than a fixed function of $M_f(|z|)$, for instance faster than a positive power of $M_f(|z|)$. This note addresses only that third question, in the natural universal sense that the comparison function is fixed in advance and is independent of f. We prove a negative answer in a strong form. For every nondecreasing function $\Phi(T) \to \infty$ there is a transcendental entire function f such that every path to infinity γ satisfies

$$\liminf_{t\to\infty} \frac{|f(\gamma(t))|}{\Phi(M_f(|\gamma(t)|))} = 0.$$

Consequently no universal comparison function, even one growing much more slowly than every positive power, can be forced in Erdős's third question. The proof is self-contained; the references are included only for provenance and context.

1 The formulation addressed here

Let f be a transcendental entire function, that is, an entire function which is not a polynomial. For $r > 0$, put

$$M_f(r) := \max_{|z|=r} |f(z)|.$$

A path to infinity is a continuous map

$$\gamma : [0, \infty) \to \mathbb{C}$$

such that

$$|\gamma(t)| \to \infty \qquad (t \to \infty).$$

Erdős's third question in Problem 514 asks whether one can force $|f(z)|$ to tend to infinity along a path faster than a fixed function of $M_f(r)$; see [1, p. 249] and the later problem list [2]. We formalize the relevant positive assertion as follows.

Problem 1 (Universal comparison-function formulation)**.** *Does there exist a function $\Psi : [T_0, \infty) \to (0, \infty)$, with $\Psi(T) \to \infty$ as $T \to \infty$, independent of f, such that every transcendental entire function f has a path to infinity γ for which*

$$\frac{|f(\gamma(t))|}{\Psi(M_f(|\gamma(t)|))} \to \infty \qquad (t \to \infty)?$$

All statements involving $\Psi(M_f(r))$ or $\Phi(M_f(r))$ are understood for sufficiently large r, so that $M_f(r) \geq T_0$. This is harmless, since $M_f(r) \to \infty$ for every nonconstant entire function.

It is no loss to restrict attention to nondecreasing comparison functions. Indeed, suppose that a not-necessarily-monotone function $\Psi : [T_0, \infty) \to (0, \infty)$, with $\Psi(T) \to \infty$, were to work. Choose $T_1 \geq T_0$ so large that $\Psi(S) \geq 1$ for all $S \geq T_1$, and on $[T_1, \infty)$ define the lower envelope

$$\Phi(T) := \inf_{S \geq T} \Psi(S).$$

Then $1 \leq \Phi \leq \Psi$, the function Φ is nondecreasing, and $\Phi(T) \to \infty$. The last assertion follows because, for each $C > 0$, we have $\Psi(S) \geq C$ for all sufficiently large S. A path with $|f(\gamma(t))|/\Psi(M_f(|\gamma(t)|)) \to \infty$ would then also have $|f(\gamma(t))|/\Phi(M_f(|\gamma(t)|)) \to \infty$. Thus it suffices to rule out nondecreasing Φ's; discarding the initial interval $[T_0, T_1)$ is irrelevant for all asymptotic statements.

We prove the following stronger negative statement.

Theorem 1. *Let*

$$\Phi : [T_0, \infty) \to (0, \infty)$$

be nondecreasing and satisfy $\Phi(T) \to \infty$ as $T \to \infty$. Then there exists a transcendental entire function f such that for every path to infinity γ,

$$\liminf_{t \to \infty} \frac{|f(\gamma(t))|}{\Phi(M_f(|\gamma(t)|))} = 0.$$

Since the quotient is nonnegative whenever it is defined, the conclusion is equivalently the following epsilon form: for every path γ, every $\eta > 0$, and every $T_* \geq 0$, there exists $t \geq T_*$, with $M_f(|\gamma(t)|) \geq T_0$, such that

$$\frac{|f(\gamma(t))|}{\Phi(M_f(|\gamma(t)|))} \leq \eta.$$

Corollary 1. *There is no universal function $\Psi(T) \to \infty$, independent of f, for which every transcendental entire function has a path to infinity along which*

$$|f(z)|/\Psi(M_f(|z|)) \to \infty.$$

In particular, the parenthetical power-type case $\Psi(T) = T^\varepsilon$, $\varepsilon > 0$, also fails.

2 Construction of the function

Fix Φ as in Theorem 1. Put

$$\varepsilon_n := (-1)^{n+1}, \qquad n = 1, 2, 3, \ldots.$$

Thus $\varepsilon_n = 1$ for odd n and $\varepsilon_n = -1$ for even n.

Choose

$$x_0 > 0$$

so large that $e^x/2 \geq T_0$ whenever $x \geq x_0$. For $x \geq x_0$, define

$$A(x) := \min\left\{\Phi(e^x/2), \frac{e^x}{4}\right\}.$$

Since both $\Phi(e^x/2)$ and $e^x/4$ tend to infinity, we have

$$A(x) \to \infty \qquad (x \to \infty).$$

We shall construct positive real numbers

$$r_n, \qquad X_n, \qquad \lambda_n, \qquad a_n$$

with $r_n \to \infty$, $X_n \to \infty$, and

$$a_n = \lambda_n r_n - X_n.$$

The entire function will be

$$f(z) := \sum_{n=1}^{\infty} \exp(-a_n + \varepsilon_n \lambda_n z).$$

The parameters are chosen so that the following off-diagonal estimates hold for all $m \neq n$:

$$\exp(-a_m + \lambda_m r_n) \leq 2^{-m-n-4} A(X_n). \tag{1}$$

We also require

$$A(X_n) \geq 2^{n+4}, \qquad X_n \geq n \log r_n + n, \qquad X_n \geq x_0, \tag{2}$$

and

$$a_n \geq n^2 + \lambda_n n. \tag{3}$$

Lemma 1. *The sequences r_n, X_n, λ_n, a_n can be chosen so that (1), (2), and (3) hold.*

Proof. We argue by induction. Suppose that the parameters have already been chosen for indices $< n$. Choose

$$r_n > \max\{r_{n-1}, n\},$$

with the convention that there is no restriction involving r_0 when $n = 1$.

The quantities

$$\exp(-a_m + \lambda_m r_n), \qquad m < n,$$

are now fixed and finite. Since $A(x) \to \infty$, we may choose X_n so large that (2) holds and, for every $m < n$,

$$\exp(-a_m + \lambda_m r_n) \leq 2^{-m-n-4} A(X_n).$$

This gives the off-diagonal estimates in which already constructed terms are evaluated at the new radius r_n.

It remains to make the new term small at all previous radii. For each $k < n$, the inequality

$$\exp(X_n - \lambda_n(r_n - r_k)) \leq 2^{-n-k-4} A(X_k)$$

will hold once λ_n is sufficiently large, because $r_n - r_k > 0$. Since there are only finitely many such k, one value of λ_n can make all these inequalities true. Increasing λ_n further if necessary, we also ensure

$$\lambda_n(r_n - n) \geq X_n + n^2,$$

which is possible because $r_n > n$. Now set

$$a_n := \lambda_n r_n - X_n.$$

Then

$$a_n = \lambda_n r_n - X_n \geq n^2 + \lambda_n n,$$

so (3) holds, and for $k < n$,

$$\exp(-a_n + \lambda_n r_k) = \exp(X_n - \lambda_n(r_n - r_k)) \leq 2^{-n-k-4} A(X_k).$$

Thus all required estimates involving index n and previous indices hold. The induction is complete. □

3 The function is transcendental entire

Let

$$f(z) = \sum_{n=1}^{\infty} \exp(-a_n + \varepsilon_n \lambda_n z)$$

be the function constructed above.

First we show that the series converges locally uniformly. Fix $R > 0$ and suppose $|z| \leq R$. If $n > R$, then by (3),

$$-a_n + \lambda_n |z| \leq -a_n + \lambda_n R \leq -a_n + \lambda_n n \leq -n^2.$$

Hence

$$|\exp(-a_n + \varepsilon_n \lambda_n z)| \leq e^{-n^2}$$

for all sufficiently large n, uniformly on $|z| \leq R$. Therefore the series converges uniformly on compact subsets of $\mathbb{C}$, and f is entire.

Next we show that f is not a polynomial. At the point

$$z_n := \varepsilon_n r_n, \qquad |z_n| = r_n,$$

the n-th term equals

$$\exp(-a_n + \varepsilon_n \lambda_n z_n) = \exp(-a_n + \lambda_n r_n) = e^{X_n}.$$

For the other terms, (1) gives

$$\sum_{m \neq n} |\exp(-a_m + \varepsilon_m \lambda_m z_n)| \leq \sum_{m \neq n} \exp(-a_m + \lambda_m r_n) \leq 2^{-n-4} A(X_n).$$

Since $A(X_n) \leq e^{X_n}/4$, this is at most $e^{X_n}/16$. Consequently,

$$M_f(r_n) \geq |f(z_n)| \geq \frac{1}{2} e^{X_n}. \tag{4}$$

Together with (2), this gives

$$M_f(r_n) \geq \frac{1}{2} e^{X_n} \geq \frac{1}{2} e^n r_n^n.$$

If f were a polynomial of degree d, then $M_f(r) = O(r^d)$, which is incompatible with the last estimate along the sequence r_n as $n \to \infty$. Thus f is transcendental entire.

4 Smallness on the wrong half-plane

Fix n, and let $|z| = r_n$. We call the half-plane

$$\varepsilon_n \operatorname{Re} z > 0$$

the correct half-plane for the n-th circle. Thus the correct half-plane is the right half-plane for odd n and the left half-plane for even n.

Suppose $|z| = r_n$ and $\varepsilon_n \operatorname{Re} z \le 0$. Then the main term satisfies

$$|\exp(-a_n + \varepsilon_n \lambda_n z)| = \exp(-a_n + \varepsilon_n \lambda_n \operatorname{Re} z) \le e^{-a_n} \le 1.$$

The other terms are bounded by

$$\sum_{m \ne n} |\exp(-a_m + \varepsilon_m \lambda_m z)| \le \sum_{m \ne n} \exp(-a_m + \lambda_m r_n) \le 2^{-n-4} A(X_n).$$

By (2), $A(X_n) \ge 2^{n+4}$, and hence

$$1 \le 2^{-n-4} A(X_n).$$

Therefore

$$|f(z)| \le 2^{-n-3} A(X_n).$$

On the other hand, from (4) and the monotonicity of Φ,

$$\Phi(M_f(r_n)) \ge \Phi(e^{X_n}/2) \ge A(X_n),$$

where $e^{X_n}/2 \ge T_0$ follows from $X_n \ge x_0$. We have proved the key estimate

$$|z| = r_n, \quad \varepsilon_n \operatorname{Re} z \le 0 \quad \Longrightarrow \quad |f(z)| \le 2^{-n-3} \Phi(M_f(r_n)). \tag{5}$$

5 Smallness on the imaginary axis

For any real y,

$$|\exp(-a_n + \varepsilon_n \lambda_n i y)| = e^{-a_n}.$$

Thus, by (3),

$$|f(iy)| \le \sum_{n=1}^{\infty} e^{-a_n} \le \sum_{n=1}^{\infty} e^{-n^2} =: B < \infty.$$

Since f is transcendental entire, in particular nonconstant, $M_f(r) \to \infty$. Hence $\Phi(M_f(r)) \to \infty$. It follows that

$$\frac{|f(iy)|}{\Phi(M_f(|y|))} \to 0 \qquad (|y| \to \infty), \tag{6}$$

where the quotient is understood for sufficiently large $|y|$, so that $M_f(|y|) \ge T_0$.

6 Every path has small values infinitely often

Let $\gamma : [0, \infty) \to \mathbb{C}$ be any path to infinity. Since $r_n \to \infty$, for every sufficiently large n the path meets the circle $|z| = r_n$. Let

$$\tau_n := \inf\{t \geq 0 : |\gamma(t)| = r_n\}$$

be the first hitting time. For all large n, τ_n is finite, $\tau_n < \tau_{n+1}$, and $\tau_n \to \infty$. These facts follow from the continuity of $|\gamma(t)|$ and the condition $|\gamma(t)| \to \infty$.

There are two cases.

First suppose that for infinitely many n,

$$\varepsilon_n \operatorname{Re} \gamma(\tau_n) \leq 0.$$

For those n, estimate (5) gives

$$\frac{|f(\gamma(\tau_n))|}{\Phi(M_f(|\gamma(\tau_n)|))} \leq 2^{-n-3} \to 0.$$

Thus the desired liminf is zero.

It remains to consider the opposite case. Then, for all sufficiently large n,

$$\varepsilon_n \operatorname{Re} \gamma(\tau_n) > 0.$$

Since ε_n changes sign with n, the real parts $\operatorname{Re} \gamma(\tau_n)$ and $\operatorname{Re} \gamma(\tau_{n+1})$ have opposite signs for all large n. By continuity of γ, there exists

$$s_n \in [\tau_n, \tau_{n+1}]$$

such that

$$\operatorname{Re} \gamma(s_n) = 0.$$

Write $\gamma(s_n) = iy_n$, where $y_n \in \mathbb{R}$. Since $s_n \geq \tau_n \to \infty$ and $|\gamma(t)| \to \infty$, we have $|y_n| = |\gamma(s_n)| \to \infty$. Hence (6) gives

$$\frac{|f(\gamma(s_n))|}{\Phi(M_f(|\gamma(s_n)|))} = \frac{|f(iy_n)|}{\Phi(M_f(|y_n|))} \to 0.$$

Again the desired liminf is zero.

Since γ was arbitrary, Theorem 1 follows.

7 Consequences and formalization note

Theorem 1 proves Corollary 1 as follows. Let $\Psi(T) \to \infty$ be any proposed fixed comparison function. If Ψ is nondecreasing, applying Theorem 1 directly with $\Phi = \Psi$ produces a transcendental entire function f for which every path to infinity has

$$\liminf_{t \to \infty} \frac{|f(\gamma(t))|}{\Psi(M_f(|\gamma(t)|))} = 0.$$

For a nonmonotone Ψ, apply Theorem 1 instead to the nondecreasing lower envelope Φ constructed in the first section. Since $\Phi \leq \Psi$ on the relevant tail, any sequence of points for which the quotient with denominator $\Phi(M_f(|z|))$ tends to zero also has the quotient with denominator $\Psi(M_f(|z|))$

tending to zero. Hence no path can make the Ψ-quotient tend to infinity. This rules out the universal comparison-function version of the third question in Erdős Problem 514.

The note does not reprove the first two questions in Problem 514; it is only a self-contained proof of the negative answer to the third, comparison-function question. The accompanying Lean 4 file `Erdos514.lean`, using `leanprover/lean4:v4.28.0` and the corresponding mathlib version, formalizes the theorem in the equivalent epsilon form. In that file the main circle-maximum theorem is `comparison_negative_sphere`; the predicate `FrequentlySmall` is the epsilon formulation above; and the circle maximum `Ms` matches $M_f(r) = \max_{|z|=r} |f(z)|$. The Lean proof first establishes a closed-disk version, `comparison_negative_closedDisk`, and then identifies it with the circle version by the maximum-modulus principle. The Lean file treats Φ as a total function $\mathbb{R} \to \mathbb{R}$, with monotonicity and positivity assumed only on $[T_0, \infty)$; this is equivalent to the partial-domain formulation used here after extending Φ arbitrarily outside the relevant tail.

References

[1] P. Erdős, *Some unsolved problems*, A Magyar Tudományos Akadémia Matematikai Kutató Intézetének Közleményei **6** (1961), 221–254.

[2] P. Erdős, *Some of my favourite problems which recently have been solved*, in *Proceedings of the International Mathematical Conference, Singapore 1981*, North-Holland Math. Stud. **74**, North-Holland, Amsterdam, 1982, 59–79.